India Semiconductor Mission: *$105B* OSAT Market Entry

A comprehensive entry framework for global OSAT leaders navigating India's high-growth semiconductor ecosystem.

Semiconductor Value Chain: India's Strategic Play

While front-end fabrication remains a long-term goal, India is rapidly becoming a global hub for OSAT and ATMP.

Raw Materials

Sand and Silica purification into Silicon Wafers (Global supply dominated by Japan/Germany).

Front-end Fab

Circuit printing on wafers. India currently lacks a commercial silicon fab, representing the "Weakest Link" in the domestic chain.

OSAT/ATMP Back-end

Assembly, Test, Mark, and Pack. This is India's fastest-growing segment, anchored by Micron and Tata investments.

PCB Assembly

Integration of chips onto boards. Strong domestic capability driven by the PLI scheme for mobile manufacturing.

End-Device

Final consumer products like smartphones and EVs. India is the world's second-largest smartphone manufacturer.

Policy

Unlocking value through the *India Semiconductor Mission* policy framework

India offers one of the world's most aggressive incentive structures to offset the capital intensity of OSAT operations.

01

Central SPECS/ATMP Scheme

Offers a fiscal support of up to 30% of project capex for assembly, testing, and advanced packaging facilities.

02

State-Level Stacking

Gujarat offers an additional 25% capex subsidy plus power at ₹1/unit; Tamil Nadu provides up to 20% grants and workforce training funds.

03

ISM 2.0 Updates (2026-27)

New ₹10,000 Cr incentive for semiconductor-grade chemicals and a 30% local content mandate for government tenders.

04

Design-Linked Incentive (DLI)

Support for Indian chip design startups, creating a domestic "customer base" for local OSAT facilities.

$105B Market Opportunity: Demand Acceleration

Market projected to grow 2.8x by 2030

Policy Certainty

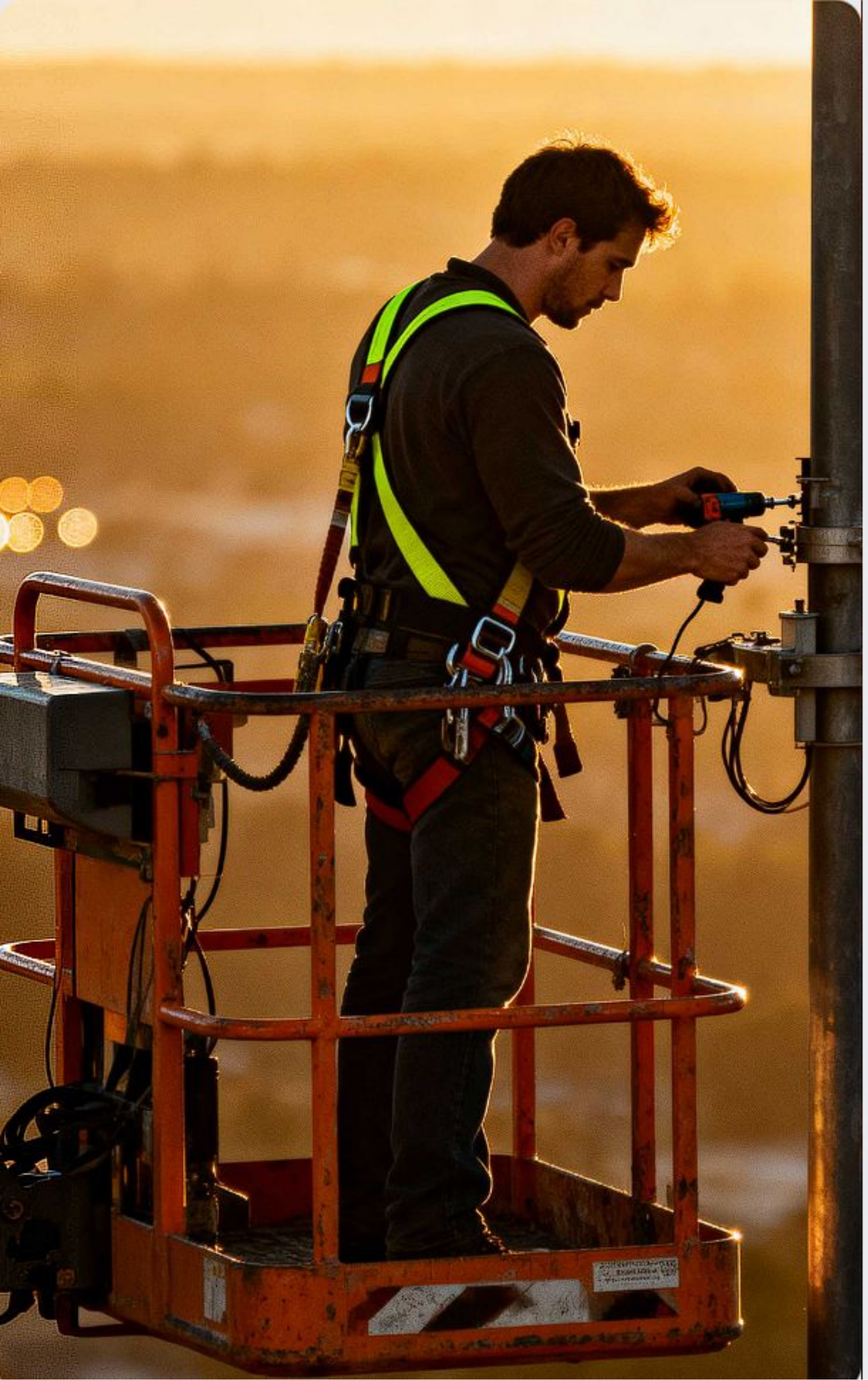
The $10B ISM corpus and stable multi-year state policies provide long-term visibility.

China+1 Strategy

Global OEMs are diversifying supply chains; India offers a reliable alternative.

Infrastructure

Dedicated clusters like GIFT City provide single-window clearances and 98.5% reliability.



Strategic segment selection: Targeting *Telecom/5G* for rapid Year 1 ramp-up

Telecom/5G Entry

Recommended for Year 1. Features 6-12 month qualification cycles and massive demand from 100,000+ new towers annually.

Auto/EV Pivot

Planned for Year 3-4. Requires AEC-Q100 certification and 18-36 month OEM qualification but offers stickier, high-ASP contracts.

Navigating the macro environment: *PESTEL* analysis

Success depends on mitigating the talent gap while leveraging strong political and legal tailwinds.

Social

50,000-engineer gap is key; 8% of grads are industry-ready.

Economic

INR volatility (±5%) is critical; Gujarat power ₹1/unit.

Political

Stable ISM policy and US CHIPS Act favor friend-shoring.

Technological

Ready for advanced packaging (SiP, flip-chip); sub-7nm goal.

Legal

IP protection improving via Trade Secrets Act 2024 and SEZ.

Environmental

Water (200-500L/chip) and waste under NGT scrutiny.

Financial Viability: The *Joint Venture* Advantage

01 **Greenfield Strategy**

Best for strategic control. Requires 74% utilization by Y3. Y5 EBITDA margins 23-25%. ~340M Capex; 6.5Y payback.

02 **JV Strategy**

Preferred for entry. Leverages partner's 300 trained technicians to accelerate revenue. ~106M Capex; 4.8Y payback.

Protecting returns against *subsidy disbursement* and utilization risks

Hurdle Rate

Project remains viable above the 16% hurdle rate unless subsidy is withdrawn.

Utilization

Greenfield IRR drops below 16% if Year 3 utilization falls under 74%.

Roadmap: *Four pillars* for India OSAT

The strategy focuses on immediate revenue generation while building a long-term automotive-grade moat.

Phase 1

JV Entry

Establish a Joint Venture in Tamil Nadu to minimize capex and leverage existing local infrastructure.

Phase 2

Segment Focus

Target Telecom/5G chips in Year 1 to ensure a 40% utilization ramp within 12 months.

Phase 3

Policy Locking

Negotiate a specific MoU with ISM to ensure milestone-linked subsidy disbursements.

Phase 4

Talent Priority

Partner with NITK and IIT Madras to train 500+ packaging engineers.